FET physics

Practice questions · Day 15 · Sat 7 Nov · pack E9

5. Practice

From your course files

WS7 Q3 (course question). Given $I_{DSS} = 6$ mA and $V_P = -4.5$ V:

a. Determine I_D at $V_{GS} = -2$ V.
b. Determine V_{GS} at $I_D = 3$ mA.

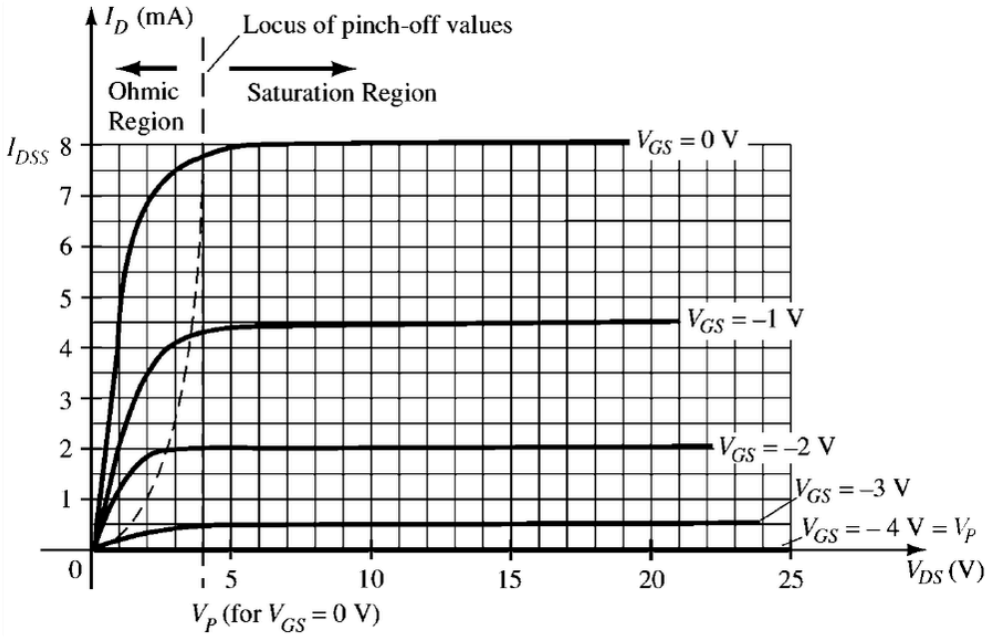
WS7 Q6 (course question). Given a Q-point of $I_{D_Q} = 3$ mA and $V_{GS} = -3$ V, determine I_{DSS} if $V_P = -6$ V.

WS7 Q5 (course question). Given $I_{DSS} = 12$ mA and $V_P = -4$ V, sketch the transfer characteristics for the JFET transistor.

WS7 Q1 (course question). Given $I_{DSS} = 12$ mA and $|V_P| = 6$ V, sketch a probable distribution of characteristic curves for the JFET.

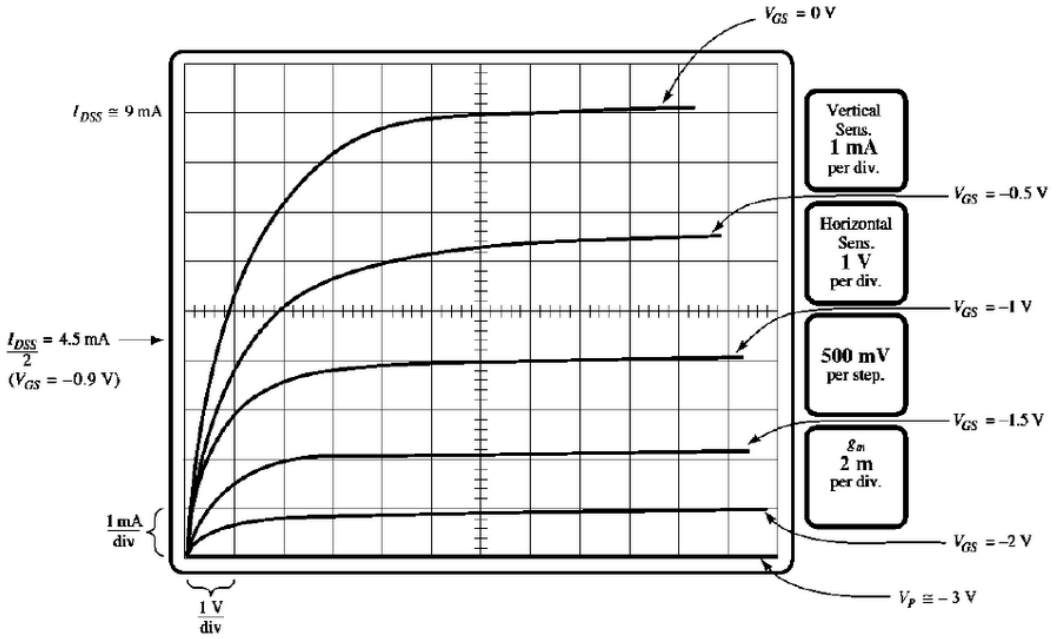
WS7 Q7 (course question). Sketch the transfer and drain characteristics of an n-channel depletion-type MOSFET with $I_{DSS} = 12$ mA and $V_P = -8$ V for a range of $V_{GS} = -V_P$ to $V_{GS} = 1$ V.

WS7 Q2 (course question). Using the characteristics of Fig. 1, determine the difference in drain current (for $V_{DS} > V_P$) between $V_{GS} = 0$ V and $V_{GS} = -1$ V.



WS7 Fig. 1 (from the workshop sheet)

WS7 Q4 (course question). Using the characteristics of Fig. 2, determine I_D at $V_{GS} = -0.7$ V and $V_{DS} = 10$ V.



WS7 Fig. 2 (from the workshop sheet)

New questions

Q1 [3 marks] A p-channel JFET has $I_{DSS} = 6$ mA and $V_P = +6$ V (Week 6 slide 13 device).

a. Find I_D at $V_{GS} = +2$ V. [1]
b. Find V_{GS} for $I_D = 1.5$ mA. [1]
c. State the direction of I_D and the sign of V_{DS}. [1]

Q2 [4 marks] An n-channel E-MOSFET datasheet gives $V_{GS(\text{Th})} = 2$ V and $I_{D(\text{on})} = 9$ mA at $V_{GS(\text{on})} = 8$ V.

a. Find k. [1]
b. Find I_D at $V_{GS} = 5$ V and at $V_{GS} = 1.5$ V. [1]
c. Find V_{GS} for $I_D = 4$ mA. [1]
d. What minimum V_{DS} keeps the device in saturation at that point? [1]

Q3 [5 marks] An n-channel depletion MOSFET has $I_{DSS} = 8$ mA and $V_P = -3$ V.

a. Find I_D at $V_{GS} = -1$ V, at -1.5 V and at $+1$ V. [3]
b. Which of these points are in depletion mode and which in enhancement mode? [1]
c. Find the exact V_{GS} for $I_D = 4$ mA, and compare it with the shorthand value. [1]

Q4 [8 marks] (chained) An n-channel JFET in saturation is measured at two bias points: $I_D = 4.5$ mA at $V_{GS} = -1$ V, and $I_D = 0.5$ mA at $V_{GS} = -3$ V.

a. Find V_P. [3]
b. Find I_{DSS}. [1]
c. Find I_D at $V_{GS} = -2$ V, and say which shorthand point this is. [1]
d. At $V_{GS} = -2$ V, find the V_{DS} at which the device enters saturation. [1]
e. In the ohmic region the channel resistance at $V_{GS} = 0$ is $r_o = 250\ \Omega$. Find r_d at $V_{GS} = -2$ V. [2]

Q5 [6 marks] A CMOS inverter runs from $V_{DD} = 5$ V. The nMOS has $V_T = +1$ V and the pMOS has $V_T = -1$ V. Its VTC gives $V_{OL} = 0.1$ V, $V_{OH} = 4.9$ V, $V_{IL} = 2.1$ V and $V_{IH} = 2.9$ V.

a. For $V_i = 0$, 2.5 V and 5 V, find each transistor's V_{GS} and whether it is on or off. Give V_o for the two logic inputs. [3]
b. Find NM_L and NM_H, and compare them with the ideal inverter. [2]
c. With $t_{PHL} = 12$ ns and $t_{PLH} = 18$ ns, find t_P. [1]

Q6 [6 marks] (hard, chained) The pMOS in a CMOS inverter has $V_{GS(\mathrm{Th})} = -1.5$ V and $I_{D(\mathrm{on})} = 5$ mA at $V_{GS(\mathrm{on})} = -6.5$ V.

a. Find k. [1]
b. Find $|I_D|$ at $V_{GS} = -4$ V, assuming saturation. [1]
c. Find V_{GS} for $|I_D| = 1.8$ mA. Show both roots and justify your choice. [2]
d. With $V_{DD} = 5$ V and $V_i = 0.5$ V, find the pMOS V_{GS} and its saturation current. State whether the nMOS ($V_T = +1.5$ V) conducts. [2]

6. Quick recall check

1. What two conditions define I_{DSS}?
2. For $V_P = -5$ V, what is I_D (as a fraction of I_{DSS}) at $V_{GS} = -2.5$ V, and what V_{GS} gives $I_{DSS}/2$?
3. Which FET type has no I_{DSS}, and what equation replaces Shockley's?
4. For a pMOS in a CMOS inverter with $V_{DD} = 5$ V and $V_i = 0$, what is V_{GS}, and is the pMOS on?
5. Write NM_L and NM_H in terms of the VTC levels.